

# Reliability Dossier — Revit Toolset & Link Infrastructure

What was broken, what it cost, and how each failure is now mitigated — permanently. Prepared by Ian Lalonde · July 28, 2026

## Executive summary

A drafter-reported symptom — “*the linked architectural model won't display*” — turned out to be **six independent defects stacked on one symptom**, and its investigation exposed a fleet-wide infrastructure disease: **3,452 broken external references across 184 active project models**, rooted in two decades of per-machine drive-letter linking and a retired file server. Every layer is now fixed, instrumented, or converted into a guided one-click repair. The toolset itself gained the reliability machinery to make this class of problem self-diagnosing: builds that verify their own freshness, tools that log what they do, and commands that explain *why* something is invisible instead of failing silently.

**6**  
stacked defects resolved

**3,452**  
dead links audited & classified

**1,770**  
relink targets located

**164**  
links healed instantly via server alias

**145**  
tools ·  
**74**  
shortcuts · Revit 2020–2027

## Part 1 — The incident: six layers, one symptom

Machine KOR-305 (C. Ford), cloud/ACC-hosted projects in Revit 2023 & 2025. Each fix was real; each revealed the next layer beneath it.

|   | What was broken                                                                      | Root cause                                                                                                                                                | Mitigation (permanent)                                                                                                                                                                                                                                     |
|---|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Machine silently ran <b>week-old tool code</b> believing it was current              | Deploy-cache defect: a locked DLL (second Revit open) blocked the update but the version stamp wrote anyway — cache marked itself current over stale code | Loader rewritten: stamp writes <b>only after every file copies</b> ; retries every launch. Plus a startup <b>freshness guard</b> : any stale machine now shouts <b>STALE BUILD</b> in its log and fails the Diagnostics self-test with the fix instruction |
| 2 | Links wouldn't return after being hidden                                             | Stranded state: the earlier category-based toggle left the Revit Links <i>category</i> hidden; the newer element-based toggle couldn't see it             | Toggle clears <b>every</b> hiding layer on the show pass: elements, category, and template-held category via temporary view properties                                                                                                                     |
| 3 | Visibility commands silently did nothing on production views                         | <b>View templates</b> govern category visibility; per-view changes are ignored without error                                                              | Template detection built into all toggles; element-level operations (which templates never control) used where possible                                                                                                                                    |
| 4 | 26 references “could not be found” in one project alone                              | Dead CAD reference paths (see Part 2 — fleet-wide epidemic)                                                                                               | <b>Data &gt; Repair Links</b> tool + fleet audit (Part 2)                                                                                                                                                                                                  |
| 5 | Cloud (ACC) arch link not loaded; disk-based repair could never find it              | Cloud links have no disk path; tooling didn't distinguish them                                                                                            | Repair Links is now <b>cloud-aware</b> : retries the ACC reload, else names the real causes (sign-in, project access, model moved). The 11 toggle lists unloaded links with the same guidance instead of staying silent                                    |
| 6 | Link loaded, category on, worksets fine — still invisible ( <b>the final cause</b> ) | <b>Per-link V/G override</b> unchecked, locked view-wide by the view template — invisible in every log, visible only on screen                            | Manually fixed (one checkbox in the template — repairs every view using it). 11 now detects this exact state and <b>names the controlling template</b> with the click-path to fix it                                                                       |

**Lesson encoded into the tools:** Revit has seven independent mechanisms that can hide a linked model, and natively reports none of them. The 11 key now checks or explains all seven — the next occurrence is a 30-second checkbox, not a week of mystery. One casualty en route (a template's custom display settings lost to an over-broad instruction) produced a standing rule: *fix visibility with the Visibility box only; never switch Display Settings off Custom.*

Part 2 — The epidemic: broken external references, fleet-wide

A server-side audit (runs on Kor-fs01 itself in ~4 minutes; rerunnable on demand) read the external-reference table of all **197 active structural models** and tested every path:

| Class                      | Count  | Meaning                                                                                                                                         | Mitigation                                                                                                                                                                                                                                                    |
|----------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Relinkable — file located  | 1,770  | File exists but moved, or was linked via a dead drive letter; audit records its <b>actual current location</b>                                  | Repair Links relinks automatically; CSV carries the target for manual cases                                                                                                                                                                                   |
| Mapped-drive links         | ~1,480 | Links made through <b>18 different per-machine drive letters</b> (P:, T:, Y:, S:, I:...) across two decades — machine-dependent by construction | Relink via UNC where live; remove where obsolete. <b>Policy: UNC-only linking</b> going forward                                                                                                                                                               |
| Retired server (BMZ era)   | 486    | \\bmz-ho-dc005 — the old company server, rebuilt as Kor-fs01; paths baked into models                                                           | <b>DNS alias resurrected the name</b> (CNAME + Kerberos SPN): 164 references healed instantly, zero model edits, 9 files restored from Archives. Path-translation config makes Repair Links rewrite them permanently; alias retires when the count reads zero |
| Foreign/consultant servers | 36     | Other firms' internal paths in received models                                                                                                  | Unfixable by design — remove-if-unused list                                                                                                                                                                                                                   |
| Dead UNC to our server     | 1      | Out of thousands: properly-made UNC links essentially never rot                                                                                 | <b>Validates the clean-infrastructure standard</b>                                                                                                                                                                                                            |

Deliverable: **KOR-Link-Audit.csv** (on the deploy share) — every broken reference, classified, with found-at targets and a worst-first project ranking for the cleanup punch list. The scanner reruns monthly so this can never silently regrow.

Part 3 — Reliability machinery now standing

- **Self-updating deploy** that can no longer lie: publish once → every machine current on next launch; a blocked update retries and reports instead of masking.
- **Self-testing suite**: Diagnostics verifies build freshness, all 145 tools, and that every configured family/image/note still resolves on the server — content moves get caught by the test, not by drafters.
- **Self-explaining failures**: every silent state found this week now speaks — wrong view type, unloaded link, closed workset, template lock — each dialog names the cause and the exact fix path.
- **Telemetry**: every family placement and repair is logged per machine; wrong behaviour is diagnosable from logs in minutes (this is how each layer above was found).
- **Server-side operations**: bulk work now executes on Kor-fs01 via remoting — never over a workstation's VPN link again.
- **Repair Links** (Data panel): one click per project — translates retired-server paths, relinks found files (RVT + CAD), retries cloud links, and reports every miss by name.

Part 4 — Open items (owned, not forgotten)

| Item                                                                                 | State                                                                                        |
|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Per-project link cleanup (relink/remove from the CSV punch list)                     | Drafter-paced; Repair Links makes each project ~10 minutes. Worst 15 ranked                  |
| BMZ alias retirement                                                                 | Automatic criterion: rescan reads zero bmz references → delete CNAME/SPN (5 min, reversible) |
| Template link-visibility sweep (the Part 1 final cause may exist in other templates) | Check during template governance review; 11 self-reports any instance found                  |
| Remaining drafter machines (Lindsay, Serban, Stephanie)                              | ~10 min each when they say go; installer handles legacy removal + shortcuts                  |
| Content consolidation of the per-year server sprawl                                  | Deliberately deferred until the pilot group signs off; server-side technique proven          |